

Joshua Goldberg

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Education

Northeastern University - Boston, MA | Khoury College of Computer Sciences

Sept 2023 - Present

Candidate for a B.S. in Computer Science, Minor in Mathematics

Expected Apr 2027

Honors: GPA 4.0/4.0 | Dean's List (4 semesters) | Honors Scholarship

Relevant Courses: Software Engineering, Programming Languages, Algorithms (Graduate), [CS7470](#) Influential Ideas in Programming Languages, Object Oriented Design, Distributed Systems

Knowledge/Expertise

Programming Languages: Java, Python, Javascript/Typescript, Lean, HTML/CSS, Rocq, OCaml, Racket, C

Applications/Systems: Git, Docker, Node.js, n8n, React, Postman, NumPy, Autograd, Prisma, Qt, IntelliJ, PyCharm

Projects

Categorical Mode - Lean 4

Jan 2026 - Present

- Implementing a DSL compiler with parsing and elaboration for reasoning about categorical structures
- Designing definitions to represent information flow and security lattices within the context of categories

OwlLean [\[source\]](#) - Lean 4

Aug 2025 - June 2026

- Developed a typechecker for [Owl](#), integrating subtyping and label corruption across 14,000+ lines of code to guarantee safety of secret data from outside adversaries
- Implemented bidirectional typing inference and automated 10+ tactics for proof generation, ensuring proper typechecking of information flow constraints
- Constructed 5+ example cryptographic programs to emulate and typecheck concurrency using state machines
- Devised performance optimizations, allowing runtimes to be 3x faster compared to unoptimized tests

Agentic Cybersecurity Bot [\[source\]](#) - Python

June 2025 - Dec 2025

- Led a team of 8 developers in the creation of a **Retrieval-Augmented Generation (RAG)** integrated Discord bot leveraging the **GPT-4 API**, and [n8n](#) workflow automation to assist 200+ students with cybersecurity coursework
- Authored and presented research at [CISSE](#), demonstrating system design with an over 80% student helpfulness rating

Osu!mania Replay Renderer [\[source\]](#) - Java

Aug 2024 - Mar 2025

- Implemented a MVC application that parses **LZMA** compressed game data and converts resulting data to MP4 format by recreating the game environment, using **7Zip** and **FFmpeg**
- Optimized object-oriented programming (OOP) design patterns within the replay renderer, decreasing average render times by over 50% and enabling smoother playback for a higher volume of user-created content

Experience

Research Assistant - Northeastern [Programming Research Laboratory](#)

Jul 2025 - Present

- Building automated tools in **Lean 4** and **Rocq** to verify security properties of cryptographic programming languages (e.g. Owl), enabling alternatives to manual proof goals
- Communicated novel implementation ideas and future plans during weekly sprints to ensure development success
- Spearheaded the formal verification of 4+ critical security properties, including non-interference and operational correctness, across 3 distinct language representations

Technical Lead - Northeastern University Electric Racing - Promoted from Team Member

Apr 2025 - June 2026

- Managing a team of 5+ developers using **React** and **Typescript**, assigning 50+ tickets and reviewing 30+ pull requests
- Built full-stack improvements for the Finance, Gantt Chart, and Calendar pages, integrating **Prisma** and **SQL filtering** and cutting management overhead and improving load times by over 50% for 200+ members across sub teams

Logic and Computation Teaching Assistant - Northeastern University - Boston, MA

Jan 2025 - May 2026

- Mentoring 100+ students weekly in preparation for proof-based exams in formal logic and property-based testing
- Maintaining autograder infrastructure in **Racket** and **OCaml**, designing systems to handle 100+ submissions weekly

Interests : Piano | Theorem Provers | American Numismatics | Programming Languages